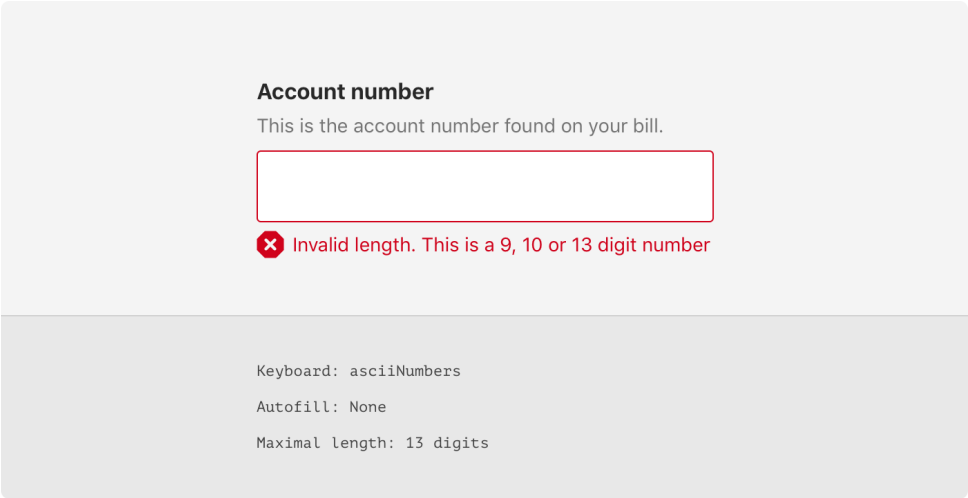
*Need help? Self-Service || Yammer || **No new spaces from Nov-23rd***

Text Input Helper

Created by Davyd Geyl, last modified by Genevieve Wang on 18 Feb 2021



Version

5

Availability

Web

Android

iOS

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- Rationale
- Usage Guidelines
- Separation of concerns
- Workflow
- Attributes
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A Text Input Helper contains data and logic to streamline entry of a particular text input type in a Text Field.

Useful Docs

- Text Input Helper Copy
- Autofill Types
- Contribute New Input Helper

Specs

- Text Input Helper Library

UI Playground

- Web Playground
- React Playground
- Android Catalogue app
- iOS Catalogue app

Rationale

Text entry of specific input types, such as email, mobile number, medicare card number, etc., is an error-prone process. Guidance and validation are required to streamline the data entry process, prevent user slips and mistakes where possible, or simplify error correction if error can't be prevented (see Usability heuristic #9).

Encapsulating guidance, validation parameters and functionality in a dedicated Text Input Helper ensures all the necessary aspects are considered in design and implementation.

Decoupling guidance and validation logic from the Text Filed allows better separation of concerns. It allows the Text Input Helpers to be easily reusable and testable.

Usage Guidelines

Each concrete Text Input Helper is designed for one specific text input type (see more in our library). For example, InputHelperBSB allows users to enter a BSB, while InputHelperMedicareNumber is designed to help with entry of Medicare card numbers.

A new concrete Text Input Helper must be created to support new input types. See here on how to contribute a new Input Helper.

Separation of concerns

View Controller, Text Field and Text Input Helper work together to capture text input from user. The responsibilities are split as follows:

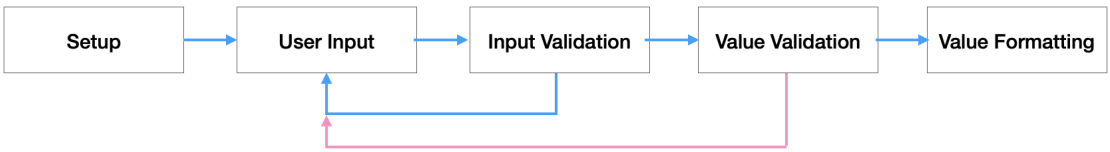
Text Field	Text Input Helper	View Controller

Text Field	Text Input Helper	View Controller
Display the Label Display the Error Message Display the Help Text Display formatted value when inactive Display Clear Content button Set Keyboard ✶ Set Capitalisation ✶ Set Autocorrection ✶ Set Autofill ✶ Set default Label ✶ Set default Help Text ✶ Have isRequired property Handle isRequired validation Have default error message for required state	Filter input based on the character set Limit input length to the defined Max Length Filter input based on specific pattern Format the value Provide the default Label ♦ Provide the default Error Messages ♦ Provide the default Help Text ♦ Provide Keyboard, Capitalisation, Autocorrection and Autofill attributes ♦	Provide the Text Field with a corresponding Text Input Helper Triggers value validation in the Text Field when needed Validates other controls

✶ the values are obtained from the provided Text Input Helper

♦ the values are available for the Text Field to use

Workflow



Setup

Text Field is provided with an instance of a specific Text Input Helper that is designed for the particular input type. Text Input Helper library contains available ones. If a required Text Input Helper does not exist it can be created by the practitioners and added to the Design System.

Text Field configures all necessary parameters, such as keyboard, capitalisation, autocorrection, autofill, etc., based on the values obtained from the provided Text Input Helper.

User Input

User will type or paste characters into the Text Field. Each time the suggested input value is passed to the Text Input Helper for validation. Based in the result the Text Field either adds the character to the Input Value or discards it.

Input Validation

The purpose of the Input Validation phase is to prevent user slips and mistakes. It comes down to the following scenarios.

- Discard characters that are not part of the desired character set if such parameter is specified.
- Limit the length to the specified Maximum Length if such parameter is specified.
- Discard characters based on specific pattern if such pattern is specified.

Errors are not displayed during this phase. The assumption is that input is not complete yet.

Copy-pasting

TextField may accept strings pasted from the pasteboard. The process is similar to the Input Validation, except it requires pre-processing of the pasted value. Validation of the proposed result with possible rejection may be needed.

1. Discard characters that are not part of the desired character set if such parameter is specified.
2. Insert the pasted string at the cursor position. If the input value already exists in the text field the pasted value is added to it.
3. If max length limitations exist trim the tail of the the result joined value.
4. Validate the input value based on specific pattern if needed. Reject the paste operation if such validation fails.

Value Validation

The purpose of Value Validation phase is to validate the input value and simplify error correction if any errors found (see Usability heuristic #9).

Assuming the input is complete, it gets validated against the set or rules specified for this particular input type.

The rules may include

- Minimum Length
- Specific pattern of characters
- Specific range of the final value

Value Formatting

When inactive and if the input value is valid the Text Field will display the formatted value. For that, the Text Field will ask the Text Input Helper to format the input value. Formatting is removed when the Text Field gets focused to allow editing of the input value.

Attributes

The following configuration attributes are used during the Setup phase.

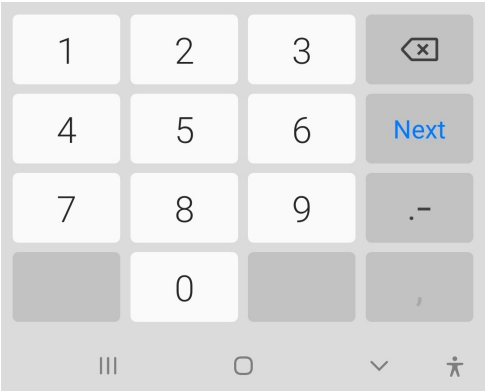
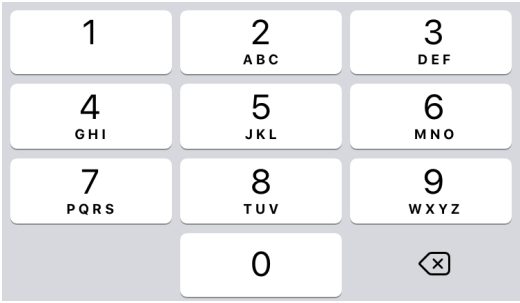
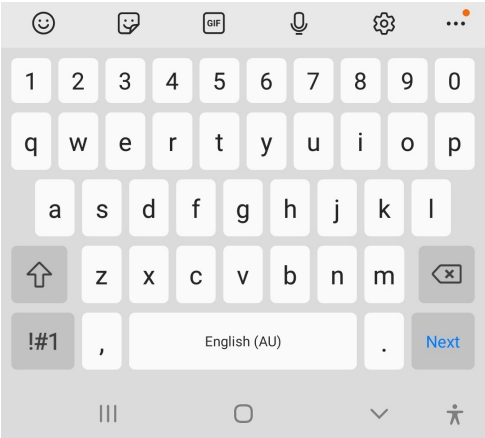
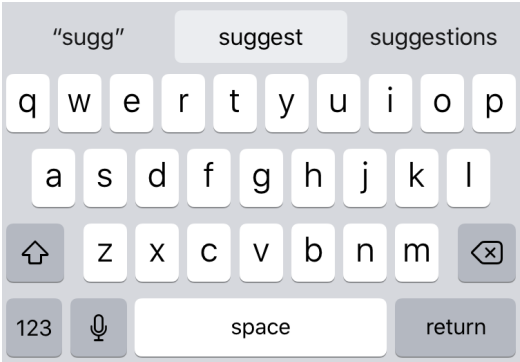
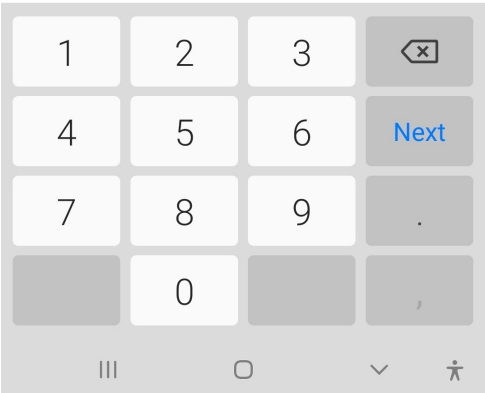
Character Set

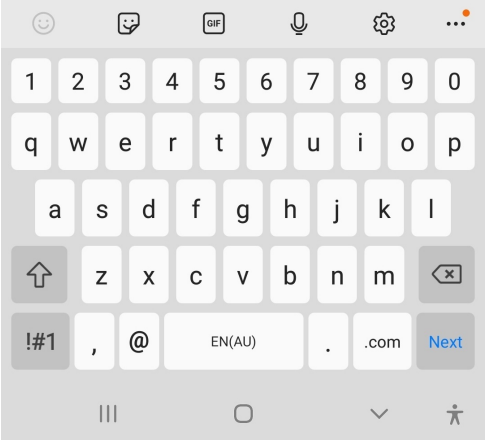
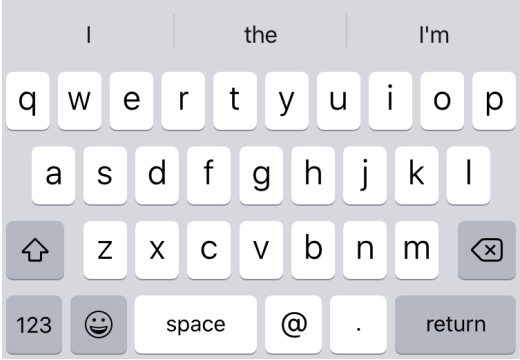
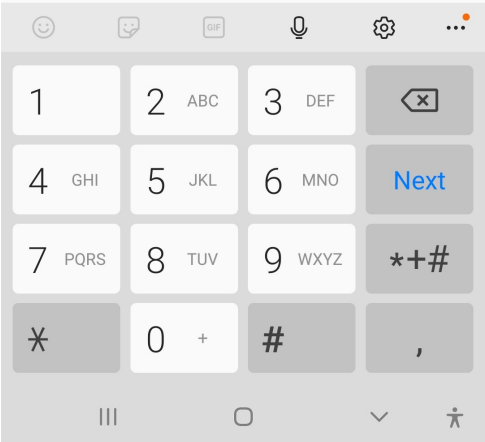
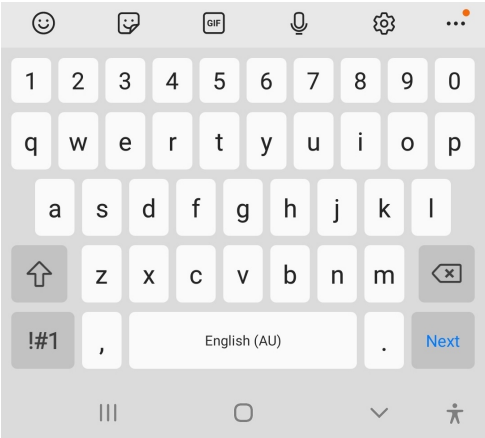
The set of unicode characters that is acceptable for a specific input type.

Name	Allowed characters Unicode	Number of characters	Description
ASCIINumbersCharacterSet	[U+0030 – U+0039]	10	ASCII only digits.
ASCIIPrintableCharacterSet	[U+0020 – U+007E]	95	ASCII printable characters.

Keyboard type

Presenting the most suitable keyboard increases chances of gathering the input without error. Several types of onscreen keyboards are available each optimised for a specific character set.

Identifier	Picture Android	Picture iOS	Description
asciiNumbers			A number pad (0-9) that will always be ASCII digits.
asciiCharacters			For entering ASCII characters
decimal			A number pad with a decimal point.

Identifier	Picture Android	Picture iOS	Description
email			Optimized for multiple email address entry.
phone			A phone pad (1-9, *, 0, #, with letters under the numbers).
name			Optimised for entering a person's name

Capitalisation

Identifier	Description	Example
none	No change to the capitalisation is done	
words	The first letter of every word is capitalised	Every Word Is Capitalised
sentences	The first letter of the input is capitalised, so is the the first letter of every new sentence.	My new phone. Love it.
allCharacters	All characters are capitalised.	CSIRO

Autocorrection

Identifier	Description
yes	Spellchecking of the input text will be done and dictionary words suggested for picking
no	No spellchecking is expected, dictionary words will not be offered

Autofill type

See Autofill Parameters for all the different content types offered by Android, iOS and Web.

Key Terms

Term	Definition
Text Field	Component of a graphical user interface that allows gathering text-based input from the user using the physical or software keyboard.
Physical Keyboard	Hardware keyboard connected to your computer or mobile device.
Software Keyboard	Component of a graphical user interface that lays out buttons mimicking physical keyboard.
Input Type	Specific piece of data with dedicated purpose, known character set, patterns and formatting. For example: Medicare card number, email, mobile phone number, BSB.
Character Set	Set of characters that is allowed to be used with the specific input type. For example, the character for of BSB input type consists only of 10 characters - numbers from 0 to 9.

Related Patterns

- Text Field
- Text Area *(Coming soon)*
- Search Field *(Coming soon)*
- Password Field *(Coming soon)*
- Date Field *(Coming soon)*

No labels

Atlassian